

upGrad

Topic 01: Set the workspace boundary

Foundations: Python, Cloud and a reproducible workspace

TOPIC 01

Set the workspace boundary

Separate the project, credentials, data and documentation from the start.

What this topic establishes

By the end of this topic, learners can

- describe the role of Python, IAM, S3 and Git in a data-science workflow;
- create a minimal local project structure that can be reproduced by a teammate; and
- distinguish credentials, code and data artefacts by where they should live.

The smallest useful data-science workspace

Local project

- versioned Python environment
- `src/` for reusable code
- `notebooks/` for exploration
- `README.md` for the next learner

AWS account boundary

- IAM role grants least-privilege access
- S3 stores durable data and artefacts
- CloudShell offers a browser-based terminal
- never commit keys or tokens to Git